

Sahil Raut

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EDUCATION

University of the West of England — BEng (Hons) Robotics | Bristol, UK

September 2021 – July 2024

Relevant coursework: Mathematical Modelling for Electronics, Robot Control Systems, Signal Theory, and Machine Vision.

- Attended the SICK Solution Hackathon 2023 in Germany.
- Served as Student Representative during Years 2 and 3.
- Served as Social Media Officer for the Indian Society.
- Participated in UWE-AI, developing an autonomous Formula Student car.

Podar International School — AS/A Levels | Mumbai, India

June 2019 – June 2021

Subjects: English, Mathematics, Physics, and Computer Science.

EXPERIENCE

NEURA Robotics — AI Robotics Software Developer | Riederich, Germany

February 2025 – Present

Develop intelligence for autonomous robotic manipulation by bridging deep-learning research and physical robot execution. Deploy AI-driven solutions that enable robots to learn and adapt to novel tasks end to end.

- Train, refine, and integrate deep-learning models, including grasp-generation systems, into production robotics platforms.
- Integrate AI software stacks with the Neuraverse cloud platform to enable communication between physical robots and cloud-based intelligence.
- Design and deploy custom robotic applications tailored to customer requirements.
- Collaborate with hardware, product, and software teams to troubleshoot complex issues and improve autonomy algorithms.
- Support the deployment of high-precision robotic manipulation solutions in real-world environments.

UWE-AI — Perception Engineer | Bristol, UK

June 2024 – February 2025

- Integrated a SICK PicoScan LiDAR sensor into an autonomous vehicle platform.
- Worked on perception capabilities for autonomous navigation and environmental sensing.

Bristol Robotics Laboratory — Robotics Engineer Intern | Bristol, UK

June 2024 – February 2025

- Modified and developed an HRI-based humanoid robot named CASTOR.
- Led the project team during the development and integration process.
- Integrated the ChatGPT API to provide voice feedback to end users.
- Migrated the existing system to a Raspberry Pi 4 and NVIDIA Jetson Nano, distributing computing and I/O responsibilities across both platforms.

UWE Bristol — Student Representative | Bristol, UK

May 2022 – July 2024

- Represented students and provided feedback to improve course offerings across the Faculty of Arts, Humanities and Education.
- Monitored trends in student feedback and worked with staff to resolve issues and implement improvements.

Islington Robotica — Robotics Engineer Intern | London, UK

May 2023 – July 2023

- Modified ROS 1 packages for the El Chapo V2 robot.
- Optimised the robot's hardware model and improved the existing mechanical build.
- Worked on 3D printing, CAD, soldering, low-voltage cabling, and electrical testing.
- Performed voltage measurements, hardware debugging, and prototype development.

SICK Sensor Intelligence — Robotics Training and Intern | Mumbai, India

August 2021 – September 2021

- Studied the operating principles and development of LiDAR, RGB colour, and infrared sensors.
- Supported sensor implementations at industrial sites, developing an understanding of Industry 4.0 applications.
- Designed and built a TurtleBot for anti-collision applications.

TECHNICAL SKILLS

Programming: C, C++, Python, VHDL

Robotics: ROS 1, ROS 2, robot control, computer vision, machine vision, visual SLAM, LiDAR, sensor integration, Arduino

Hardware and Prototyping: Electronics hardware design, PCB soldering and debugging, prototyping, 3D printing, 3D CAD, hardware testing

Software and Tools: Git, GitHub, Kubernetes, Gazebo, MoveIt, MATLAB, Fusion 360, SolidWorks, Choregraphe, Raspbian

Professional: Project management, team leadership, communication, user-requirements analysis, research

PROJECTS

Automating Trash Sorting — Project | UWE Bristol

October 2023

- Developed an automated waste-sorting system using a SICK Visionary-S 3D stereo camera, YOLOv8-based classification, Azure Custom Vision, and an ABB GoFa CRB 15000 collaborative robot.
- Reduced manual involvement in waste sorting through automated identification and categorisation of different waste types.
- Improved the speed, consistency, and precision of the recycling workflow.

Robotic Arm Automation and Control System — Project | UWE Bristol

January 2024

- Developed an automated robotic-arm system using communication protocols and servo-motor control.
- Designed a user interface with an OLED display and RGB LED indicators for real-time feedback and system monitoring.
- Implemented precise movement algorithms, manual control, and an emergency-stop feature.

HRIT Real-Time Adventure Game Using NAO Robot — Project | UWE Bristol

January 2024

- Developed a human–robot interaction system combining large language models with robotics to create an interactive, real-time adventure game.
- Addressed challenges involving voice recognition, Choregraphe programming, and hardware limitations.
- Explored the use of generative AI in entertainment and human–robot interaction.

CASTOR Humanoid — Project | Bristol Robotics Laboratory

January 2024 – June 2024

- Led the development of the CASTOR humanoid robot, including hardware assembly and software configuration.
- Integrated facial-expression synchronisation using an OpenMV H7 Plus camera.
- Developed Python scripts to synchronise facial expressions and improve user engagement.

SICK Sensors Hackathon 2023 — Project | Germany

2023

- Designed and developed sensor-based robotic solutions during the SICK Solution Hackathon.
- Applied LiDAR and other sensing technologies to autonomous robotics challenges.

Micromouse — Project | UWE Bristol

May 2023

- Led the project through CAD design, PCB design using Eagle, infrared sensor construction, and motor testing.
- Programmed a maze-search algorithm that enabled the robot to navigate obstacles and maze walls.

ACTIVITIES AND INTERESTS

Sports: Football, table tennis, badminton, swimming, and chess.

Interests: Formula 1, music, cinematography, gaming, video editing, and content creation.